

capabilities such as ACID transactions, schema evolution, and time-travel queries, the lakehouse offers a single platform for BI, data science, and real-time analytics.

What is a data lakehouse?

A data lakehouse integrates the best practices of data lakes and data warehouses, filling the gap between scalable, flexible storage and transactional, dependable analytics. It provides an integrated platform where organisations can oversee the entire lifecycle of the data — from ingestion and processing to analytics and machine learning.

Traditional data lakes are planned for raw, bulk storage but do not include the capabilities necessary for enterprise-level analytics, including schema enforcement, data versioning, and ACID guarantees. Data warehouses have these features but are expensive, inflexible, and usually associated with proprietary vendors.

The lakehouse resolves this trade-off by putting warehouse-like capability into open data structures (such as Parquet and ORC), while keeping the scalability and economics of object storage systems (such as S3, HDFS, or GCS).

Apache Iceberg: Modern table format for the lakehouse

With the changing paradigm of data platforms, storing data files in a data lake is not enough anymore. Organisations need transactional consistency, schema evolution, time travel, and performance at scale with open file formats and cloud-native principles intact. This is where Apache Iceberg comes into play.

Originally developed by Netflix and now an Apache top-level project, Iceberg rewrites the rules for table management in big data lakes. Unlike Hive-style tables, which are based on directory hierarchies and static metadata, Iceberg provides a highly optimised metadata layer that supports

correct operation on huge datasets.

Iceberg is built to address the problems experienced by data analysts and engineers — the lack of safe evolution of schemas, variability in query results between engines, and performance bottlenecks due to poor partitioning. With Iceberg, these issues are solved with a modular design that separates storage from metadata so that it allows for authentic multi-engine interoperability between Spark, Trino, Flink, and others.

Versioned snapshots in the table format enable time travel, schema evolution, and even partition evolution so that data organisations can change over time without having to rewrite the entire dataset. These features make your data lake a trusted, enterprise-quality analytical system without vendor lock-in.

Trino: Federated SQL engine for the modern lakehouse

In the era of distributed, multi-source data, organisations need a fast, scalable SQL engine that can unify access without centralising storage. Trino (formerly Presto SQL) is designed to meet this need, and it plays a central role in modern lakehouse architectures.

Here's what makes Trino exceptional.

- ▶ **Federated query engine**
 - Trino supports federated queries across diverse source object stores, warehouses, RDBMS, NoSQL, and streams.
 - It eliminates the need to move data; instead, it pushes down queries to wherever the data lives.
 - Ideal for data mesh environments where different teams own different data domains.
- ▶ **Massively parallel and cloud native**
 - Trino is a distributed MPP (massively parallel processing) engine.
 - It separates compute from storage, enabling elastic scalability.
 - Executes queries using ANSI

SQL, making it familiar to data analysts and engineers.

- ▶ **Broad connector ecosystem**

- It supports a wide range of backends like Apache Iceberg, Hive, Delta Lake, Kafka, Elasticsearch, MongoDB, PostgreSQL, MySQL, and many others.
- Perfect for hybrid and multi-cloud architectures.

- ▶ **Optimised for Iceberg**

- Offers native support for Apache Iceberg with full access to:
 - Time travel and snapshots
 - Hidden partitioning and schema evolution
 - Metadata-based planning for fast query execution
- Trino + Iceberg equals a truly open lakehouse stack.

- ▶ **Enterprise-ready and extensible**

- Integrates easily with BI tools (e.g., Tableau, Superset, Power BI).
- Plugs into data catalogues (e.g., Glue, Hive, Nessie) for metadata management.
- Supports fine-grained access control and query observability.

Building a lakehouse with Apache Iceberg and Trino

Designing a modern data platform no longer means choosing between scale and structure. The lakehouse architecture, powered by Apache Iceberg and Trino, allows organisations to build an open, high-performance data system that supports everything from BI to AI without vendor lock-in.

At the core of this architecture is a simple but powerful idea: decouple storage, metadata, and compute. Each layer is built using open technologies that work together seamlessly.

Best practices and recommendations

To ensure your data lakehouse remains performant, scalable, and manageable, it's important to follow architectural